MoPac+: Multimodal **Mo**deling of the **Pa**vement **C**racks using Hard-Example Mining and Context-Aware Feature Aggregation⋆

Debojyoti Biswas[a,*,1], Andrew Scouten[a,1], Haitao Gong[a,2], Feng Wang[a,2] and Jelena Tešić[a,1]

[a]*Texas State University, 601 University Ave, San Marcos 78666, Texas, U.S.A.*

ABSTRACT

Automated pavement distress (crack) detection through image analysis remains critical for pavement condition evaluation, maintenance planning, and asset management. While deep learning has significantly advanced pavement distress detection, current datasets and methodologies predominantly focus on common crack types (e.g., longitudinal/transverse), leaving rare distress patterns and fine-grained categorizations inadequately addressed. Existing approaches further demonstrate limitations in handling irregular crack morphologies and data-scarce scenarios. We introduce *MoPac+*, a novel anchor-free multimodal framework with Intensity and Range imagery (I+R) that integrates context-aware feature aggregation, adaptive hard-example mining, and loss reweighting, designed explicitly for robust localization and classification of pavement deterioration. Our methodology addresses three key challenges: (1) uncontrolled crack geometry with significant intra-class size variation, addressed through anchor-free detection modules, and (2) grayscale image limitations mitigated by introducing multimodal training and a series of preprocessing techniques that enhance distressed regions. (3) improves the detection efficiency of ambiguous and rare classes by implementing context-aware feature aggregation and multimodal feature fusion. Comprehensive experiments demonstrate substantial performance gains, achieving a **7.5**% relative improvement in mean average precision (mAP) over baseline YOLOv8 models while maintaining computational efficiency. These advancements establish new state-of-the-art benchmarks for imbalanced pavement distress recognition in resource-constrained environments.

1. Introduction

Efficient pavement management is fundamental to maintaining highway infrastructure in good condition and ensuring safety to the driving public, as pavement structures are continually exposed to environmental weathering and heavy traffic loads that accelerate deterioration. Early and accurate detection of pavement cracks and other distresses is vital for extending service life and optimizing maintenance and rehabilitation (M&R) activities. The TxDOT [1] maintains an extensive roadway network spanning over 703 thousand lane miles and supporting an annual travel volume of nearly 301 billion miles. Simple arithmetic shows that each average lane-mile interacts with vehicles more than 428,000 times per year, underscoring the need for robust automated pavement crack detection systems that enable timely, cost-effective maintenance. Automated pavement crack detection typically relies on advanced imaging sensors and computer vision algorithms to identify and characterize surface defects. However, accurate detection remains challenging due to the inherent complexity and variability of pavement distress types, materials, pavement types, and surface texture characteristics. Pavement surfaces may exhibit a broad spectrum of defects, ranging from minor hairline cracks to severe structural failures, and each pavement type, such as asphalt or concrete, presents unique visual characteristics. Sensor heterogeneity, variations in image quality, intra-class diversity, and inter-class similarities further complicate detection, often limiting the generalizability and reliability of automated solutions.

Recent advancements in deep learning [2, 3, 4], particularly convolutional neural networks (CNNs), have substantially improved the accuracy of pavement crack detection and classification. State-of-the-art object detectors, including single-stage models like YOLO [5] and end-to-end transformer-based models such as DETR [6], as well as two-stage frameworks such as Faster R-CNN and Mask R-CNN [7], have demonstrated strong performance on well-defined and prominent pavement damage. Despite these advances, existing approaches often struggle with detecting small, irregular, or rare distress types and typically provide only coarse localization. The challenges are exacerbated by the scarcity of comprehensive, annotated datasets and the lack of standardized benchmarks, which hinder the development, evaluation, and comparison of new methods, as evidenced in the following research [8, 9, 10, 11, 12].

The main contributions of this paper are: (i) introduction of an efficient anchor-free multimodal crack detection architecture with context-aware feature aggregation; (ii) the development of advanced data augmentation and enhancement strategies to improve dataset quality and model generalizability; (iii) implementation of hard-example mining with

⋆This research has been partially funded by the TxDOT Project #0-7150 and NSF Project 2213694

*Corresponding author

ubq3@txstate.edu (D. Biswas); yzb2@txstate.edu (A. Scouten); h_g153@txstate.edu (H. Gong); f_w34@txstate.edu (F. Wang); jtesic@txstate.edu (J. Tešić)

ORCID(s): 0000-0002-8842-0207 (D. Biswas); 0000-0001-0000-0000 (A. Scouten); 0000-0001-5818-8411 (H. Gong); 0000-0002-1528-9711 (F. Wang); 0000-0002-9972-9760 (J. Tešić)

[1]Computer Science

[2]Civil Engineering

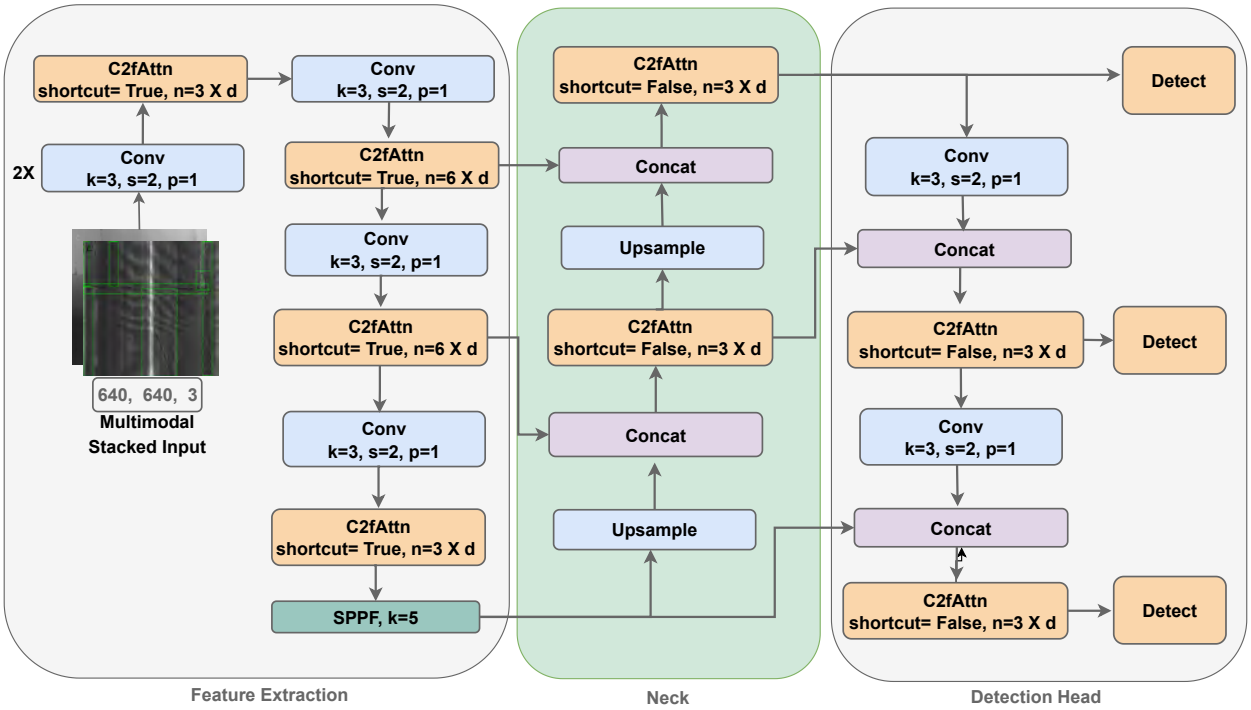


Figure 1: MoPac+: The proposed architecture, abstractly similar to YOLOv8, with detailed implementations of the Feature Extraction (Backbone), Neck, and the Detection Head. The Detection Head comprises three detection modules that enable multi-scale prediction. Here, the **yellow modules** are newly proposed in the MoPac+ framework and are illustrated in detail later.

customized focal loss function and weight-balancing mechanisms tailored to the unique characteristics of pavement distress datasets. Specifically, we propose *MoPac+*, **Mo**deling **Pa**vement **c**racks, a novel anchor-free multimodal detection framework designed to robustly localize and classify both common and rare pavement distress classes, even in data-constrained settings. Figure 1 illustrates how *MoPac+* integrates multimodal input: intensity and range images, to overcome poor performance under poor illumination and limited semantic properties from gray-scale intensity images. Next, *MoPac+* integrates a multi-stage feature-extraction backbone, a context-aware neck for multi-scale feature aggregation, and an anchor-free detection head for precise bounding-box regression and classification. The anchor-free architecture is particularly well-suited to the diverse, uncontrolled geometries of pavement cracks, thereby overcoming the limitations of traditional anchor-based detectors. To further enhance detection performance, especially for rare and subtle distress types, we introduce targeted data augmentation, feature enhancement with a context-guided C2f module, and a loss reweighting strategy that combines focal loss and weight balancing to mitigate class imbalance and prioritize complex examples. The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 details the proposed *MoPac+* methodology; Section 4 describes the dataset creation and characteristics; Section 5 presents experimental results and analysis; and Section 6 concludes with a summary of findings and future research directions.

2. Related Work

Liu et al. [13] pioneered early pavement crack detection methods that operated under the assumption that crack pixels are darker than their surroundings, and introduced thresholding algorithms, such as the adjacent difference histogram approach, that effectively distinguished crack regions from non-crack areas by maximizing pixel contrast. It further advanced detection using a connected-domain algorithm with directional segmentation, which leveraged spatial relationships to improve accuracy. Despite their computational efficiency, these threshold-based techniques are susceptible to image noise and often yield false positives because they rely solely on grayscale intensity. Next, [14] reduced false positives by analyzing gray values along object contours. In contrast, Li et al. [15] segmented images into overlapping sub-regions to better detect small-scale and complex cracks. However, these traditional methods struggle with large-scale distress and cluttered backgrounds.

Redmon et al. [5] marked a significant shift in the field with the introduction of the YOLO (You Only Look Once) family of models, which enabled real-time crack detection by predicting bounding boxes and class labels directly from input images. Successive iterations, notably YOLOv5, have incorporated multi-scale prediction, architectural enhancements, and advanced data augmentation, establishing them as leading baselines for pavement analysis. Recent studies, such as those by [16], [17], and [18], have further refined YOLO-based models by integrating transformer modules to

capture long-range dependencies and contextual information. Wu et al. [19] proposed a lightweight crossed-feature pyramid network and an improved loss function, achieving a mean average precision (mAP) of 69.3% on challenging datasets. These advancements have enabled robust detection under varying conditions, including shadows and occlusions. The widespread adoption of YOLOv5 in works such as [20], [21], and [22] underscores its effectiveness for high-precision, real-time pavement crack analysis. On the other hand, Cao et al. [23] propose a Coarse-to-fine-grained Deformable Transformer (CF-DETR) for lightweight, end-to-end object detection, focusing on small objects. However, most research has focused on common crack types, such as longitudinal and transverse cracks, leading to class imbalance and limited generalizability.

Recent research [24, 25] introduced two-stage detectors, such as the proposed separation-combination strategy and CrackDiffusion, for pavement distress detection. These models utilize region proposal networks and multi-stage refinement to enhance detection accuracy, particularly in complex scenes. Comparative analyses by Nguyen et al. [26] have demonstrated that Faster R-CNN outperforms other deep learning methods for crack detection on high-resolution datasets. Further innovations include multi-layer feature fusion networks [27], sensitivity detection networks for complex backgrounds [28], and transformer-based segmentation for real-time pixel-level analysis [29]. While these approaches achieve high accuracy on standard datasets, their performance often degrades when detecting small or rare distress types or when applied to non-standard image resolutions.

As crack pavement is vulnerable to weather and illumination, there is a separate line of work to improve performance under challenging conditions using multimodal data. For example, RGB-only crack segmentation approaches can be sensitive to changes in field conditions and often generalize poorly due to limited training data. To mitigate these challenges, Liu et al. [30] propose a multimodal crack segmentation framework (FC_SAM) built on fine-tuned large-scale vision models. Their pipeline first performs automatic registration to align infrared imagery with RGB data, thereby enabling the fusion of complementary cues for improved crack localisation. They then apply two parameter-efficient fine-tuning strategies to adapt the foundation model effectively. In a related direction, to strengthen detection when targets exhibit weak or blurry visual cues (e.g., pavement cracks), [31] introduces a Multi-scale Feature Enhancement (MFE) module that combines adaptive learning with multi-scale contextual fusion to enrich feature representations. Meanwhile, Zhai et al. [32] boost recall by treating bounding-box coordinates and mask outputs as paired multimodal descriptors for the same crack region, and develop a one-stage Multimodal Feature Fusion Network (MFFNet) that improves both detection accuracy and segmentation completeness. Although significant progress has been made, both single-stage and two-stage detection approaches still face challenges in pavement crack detection. Most existing research focuses on standard recurrent distress classes, such as longitudinal and transverse cracks, and rare classes or more detailed, specific categorizations are often neglected.

This imbalance leads models to be proficient at identifying common distresses but to fail to accurately detect rarer and critical pavement issues, thereby limiting their practical application. Hence, model improvements remain essential to supporting the durability and maintenance of transportation infrastructure worldwide.

3. MoPac+ AI Methodology

The growing importance of AI in pavement condition assessment necessitates the development of a practical yet robust Deep Neural Network (DNN)- based distress detection model capable of handling real-world data. Earlier, [33] introduced the *MoPac* skeleton baseline for practical deployment in road maintenance and safety assessment, addressing class imbalance in the dataset that biases detection. To do this, *MoPac* used YOLOv5 and YOLOv8 as feature extractors and experimented with different SVM and multi-scale detection heads. The experiments from *MoPac* showed high Average Precision (AP) in recurrent labels (AP=0.96 for Joint Cracking, with 10,283 annotations) and very low Average Precision (AP) in rare labels (AP=0.02 for Failed Concrete Patching, with 27 annotations) using an unmodified YOLOv8. The *MoPac* implementation achieved very marginal improvements for rare labels at the cost of detection accuracy of recurrent labels. At the same time, the unmodified YOLOv8 experiments show that the recurrent labels are overtaking the rare labels during training. *MoPac* 's experiments highlight the need for more robust methods, data collection, and annotation validation.

We propose the *MoPac+* architecture, an improvement over the baseline *MoPac* implementation. *MoPac+* still leverages the You Only Look Once (YOLOv8) deep learning algorithm while also introducing specific enhancements tailored to the distinctive characteristics of pavement imagery classes. The overall architecture of YOLOv8 [34] is presented in Figure 2, which shows that the YOLOv8 model does not have an explicit Region Proposal Network (RPN). The model comprises three modules: the Backbone (for deep Feature Extraction), the Neck (FPN + PAN + C2f), and the Detection Head for classification. **MoPac+** utilizes several reusable blocks or modules from YOLOv8, which are combinations of DNN layers. Such as *Convolutional* (*Conv.*) block, which utilizes a Conv2d layer with a 3×3 kernel, a stride of two, and a padding of one, followed by a BatchNorm2d layer.

The Deep Feature Extractor: As illustrated in Figure 1, the feature extractor takes multimodal inputs: Intensity and Range images (I+R), which are resized and augmented during the data preprocessing step. Next, the preprocessed input images are stacked channel-wise to create a single multimodal image. For feature extraction, MoPac+ employs a customized *CSP-DarkNet53* [35] network as the Backbone due to its demonstrated efficacy in detecting small objects

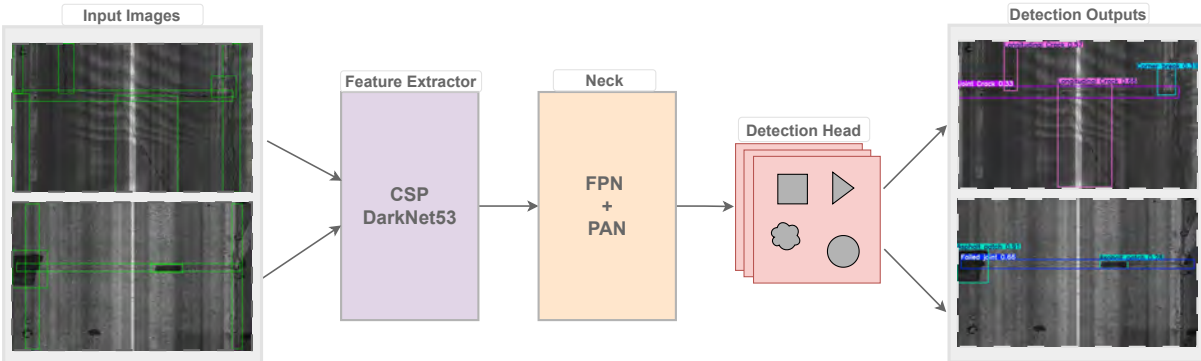


Figure 2: YOLOv8 Baseline Architecture with three modules: (1) Feature Extractor; (2) the Neck, and (3) Detection Head. Here, the input images are an intensity image and a range image of the same scene.

[36] and preserving spatial information across deep convolutional layers [37, 38]. Figure 1 illustrates the proposed architecture, combining sequential *Conv.* and *C2fAttn* modules across the backbone to help guarantee stability in feature extraction and enable robust multi-scale representation learning [34].

C2fAttn: is *CSP bottleneck with context-aware feature aggregation*, illustrated in Figure 3, extends the standard C2f design by integrating lightweight attention driven by a *feature-derived global context guide* from the same feature map being processed. Given an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, C2fAttn follows the CSP-style split–transform–merge strategy to promote efficient gradient flow, reduce computational redundancy, and improve feature reuse: $\mathbf{X}$ is first projected and split into two partitions, where one branch is forwarded. The other is processed by a stack of n *Bottleneck* modules [38]. To inject global context, we derive a guide vector via global average pooling followed by a learnable projection,

$$\mathbf{g} = \phi(\mathrm{GAP}(\mathbf{X})), \qquad \mathrm{GAP}(\mathbf{X}) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{X}_{:,:,i,j}, \quad (1)$$

where $\mathbf{g} \in \mathbb{R}^{B\times C_g}$ and $\phi(\cdot)$ denotes a lightweight MLP (or 1×1 projection). This context vector conditions the aggregation by producing channel-wise gates that modulate the same-stage features,

$$\boldsymbol{\alpha} = \sigma(\mathbf{W}\mathbf{g}), \qquad \widetilde{\mathbf{X}} = \mathbf{X}\odot\boldsymbol{\alpha}, \quad (2)$$

where $\boldsymbol{\alpha} \in \mathbb{R}^{B\times C\times 1\times 1}$ is broadcast across spatial dimensions, $\mathbf{W}$ is a learnable projection, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ denotes channel-wise multiplication. The gated features $\widetilde{\mathbf{X}}$ are then propagated through the C2f pathway and concatenated with the bypass branch under the CSP formulation, enabling context-aware feature aggregation with minimal overhead. This self-guided design is particularly effective for pavement crack detection, where cracks are often *thin*, *low-contrast*, and easily confused with sealed joints, stains, or surface texture. The global context extracted from $\mathbf{X}$ amplifies crack-consistent channels while suppressing background responses, improving crack continuity, and reducing missed detections on ambiguous or fragmented crack patterns.

Finally, the *Spatial Pyramid Pooling - Fast* (*SPPF*) module builds upon the traditional Spatial Pyramid Pooling (SPP) structure, refining the concept by applying a series of fixed-size max pooling operations using a 5×5 kernel with a stride of one on a single feature map. Unlike the original SPP [39], where pooling operations are performed in parallel, SPPF stacks pooling into sequential operations, resulting in faster inference and reduced memory consumption while still aggregating spatial context across multiple receptive fields. After the pooling operations, the resulting feature maps are concatenated along the channel dimension and subsequently passed through a 1×1 convolutional layer. This step reduces dimensionality and promotes effective feature fusion. By encoding object information at various scales, the SPPF module significantly improves detection performance, particularly for objects exhibiting substantial size variability. The SPPF module achieves an optimal balance between accuracy and efficiency, making it an essential component of our crack detection framework. This Backbone configuration ensures consistent feature map dimensions while also hierarchically capturing fine-grained patterns and structural details critical for crack detection.

The Neck in YOLOv8 is composed of the combination of a Feature Pyramid Network (FPN) and a Path Aggregation Network (PAN). Broadly, the *FPN* helps the YOLOv8 model to perform multi-scale feature mapping for spatial details and rich semantic context. On the other hand, *PAN* helps localize object proposals more accurately by accounting for the object's scale and position. The *FPN* (Top-Down) module injects semantic information from deep layers into finer-resolution features for small object detection using the *C2fAttn*, *Concat*, and *Upsample* operations. On the other hand, *PAN* (Bottom-Up) passes spatial detail back down to deeper layers to improve context and large-object detection, using *Downsampling* via strides and *C2fAttn* operations.

The Detection Head in YOLOv8 has both regression and classification branches, as illustrated in Figure 5. It plays a crucial role in generating prediction data, such as object

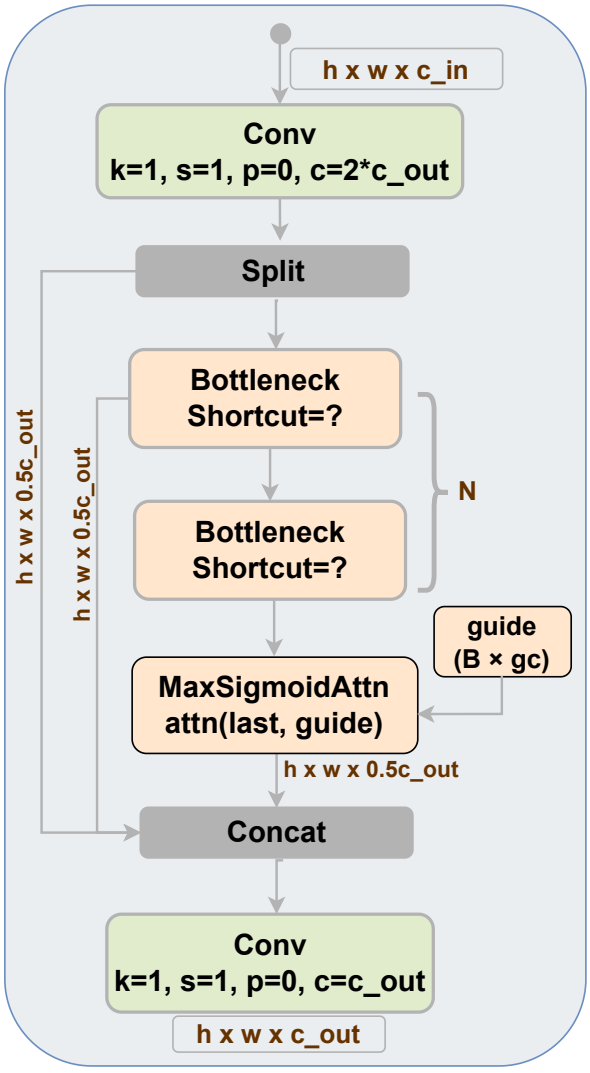


Figure 3: The *C2fAttn* block, notably consisting of N sequentially repeated Bottleneck modules followed by a *MaxSigmoidAttn* block. Here, k, s, p, and c are respectively the kernel size, stride, padding, and output channel dimension during the convolution operation. The guide is a learnable context vector derived from the features that indicates to the attention block what to emphasize in the feature map.

Figure 4: A sample of 2D Intensity, surface-level detail, (left) and 3D Range, depth, (right) pavement surface images with an Asphalt Patch annotated.

localization and classification, by leveraging an anchor-free mechanism. Unlike traditional anchor-based methods that rely on predefined anchor boxes, YOLOv8 considers each spatial location (or grid cell) in the output feature map as a potential object center—given a feature map of size $H \times W \times C$ (H=Height, W=Width and C=Output Channel), each grid cell outputs a fixed set of predictions: the objectness score (p_o), the class probabilities for each category (p_c), and the bounding box offsets (x, y, w, h). The coordinates (x, y) represent the center of the predicted object relative to the corresponding grid cell, while w and h are the width and height of the expected object. We can then formulate the bounding box coordinates in Equation 3 [34].

$$x = \sigma(\hat{x}) + c_x, \qquad y = \sigma(\hat{y}) + c_y,$$
$$w = e^{\hat{w}}, \qquad h = e^{\hat{h}} \tag{3}$$

The $(\hat{x}, \hat{y}, \hat{w}, \hat{h})$ are the raw outputs of the model, σ is the sigmoid activation that ensures the localization within the cell, and (c_x, c_y) denote the top-left offset of the grid cell in the image space. As shown in Figure 1, we perform predictions at different scales (e.g., P3, P4, P5), enabling effective multi-scale detection. The final predictions undergo Non-Maximum Suppression (NMS) to eliminate redundant bounding boxes, retaining only the most confident detections. After we have the classification score for each category from p_c, we make the final class prediction using the *argmax* function on p_c. This anchor-free approach simplifies training and reduces hyperparameter tuning, making it highly suitable for dense, varied object detection tasks such as pavement distress analysis.

$$\mathcal{FL} = \sum_{p,y} \mathcal{FL}(p, y)$$
$$\mathcal{FL}(p, y) = -\sum_{c=1}^{C} \alpha_c \, (1 - p_c)^{\gamma} \, y_c \, \log(p_c), \quad \alpha_c = \frac{1}{N_c} \tag{4}$$

The Loss Function in the *MoPac+* architecture addresses class imbalance through (i) hard-example mining using Focal Loss [40] to replace standard Cross-Entropy Loss, and (ii) class-weighted loss calculation where weight $\alpha_c = \frac{1}{N_c}$ (inverse of instance count per class). The class weights α_c are normalized to prevent gradient explosion. The focal loss ($\mathcal{FL}$) in Equation 4 [40] is the sum of the weighted cross-entropy loss function over all classes c. The p_c represents the model's estimated probability for class c, and γ is the focusing parameter.

$$\mathcal{L}(y, \hat{y}) = W_c * \mathcal{FL} + W_r * \mathcal{RL}, \quad \text{where:}$$
$$\mathcal{RL} = \frac{1}{N} \sum_{i=1}^{N} \left(\|\mathbf{b}_i(x, y, w, h) - \mathbf{b}_i(\hat{x}, \hat{y}, \hat{w}, \hat{h})\|_1 \right) \tag{5}$$

The overall loss function ($\mathcal{L}$) is defined in Equation 5 as the sum of the weighted classification elements with

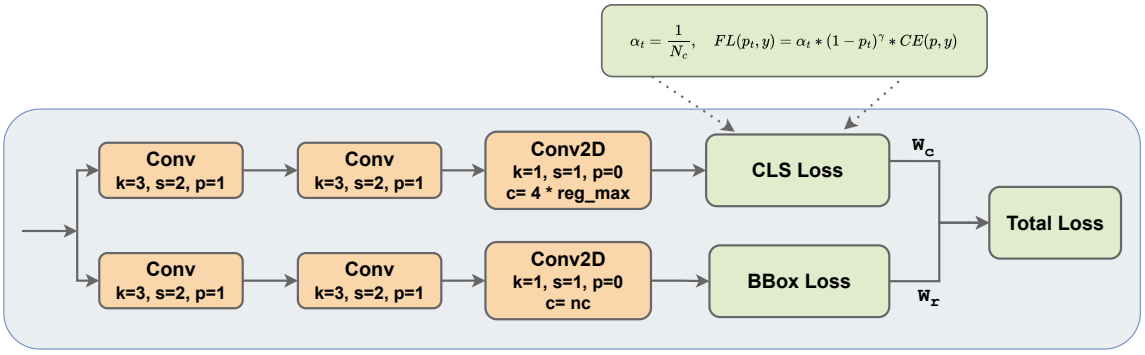


Figure 5: The *Detection Head* has two branches. 1) Classification branch, 2) Regression branch. This module performs weight balancing and hard-example mining.

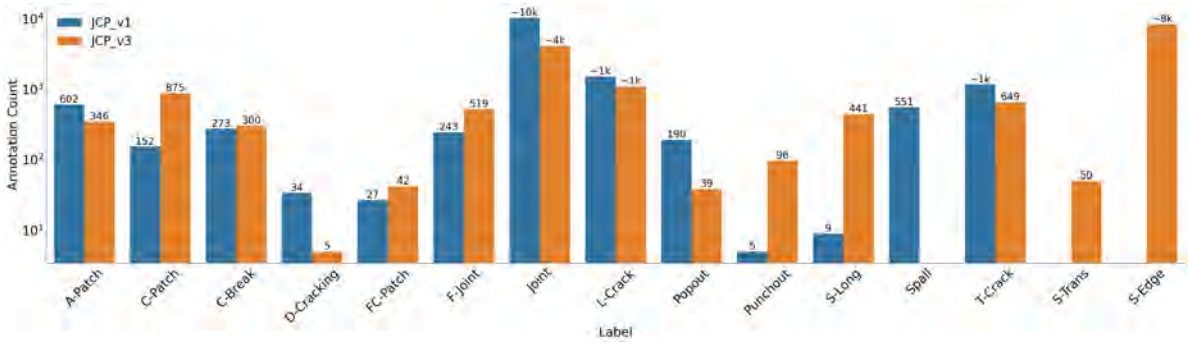


Figure 6: Annotation class counts by JCP dataset version, before any data augmentation, where MoPac uses JCPv1, JCPv2 is an intermediary version, and MoPac+ uses JCPv3.

W_c, and regression loss elements weighted with W_r. The regression task for crack detection is less complex than the classification task, thus $W_r << W_c$ in the implementation. Not that the Regression Loss ($\mathcal{RL}$) is the sum of all offsets between the ground truth bounding box $\mathbf{b}_i(x, y, w, h)$ and the predicted box $\mathbf{b}_i(\hat{x}, \hat{y}, \hat{w}, \hat{h})$ for each instance i in Equation 5.

4. MoPac+ AI Ready Datasheet

The growing importance of AI in pavement condition assessment necessitates systematic approaches to dataset creation and documentation. [41] pioneered the concept of "datasheets for datasets" to standardize transparency in machine learning data pipelines, emphasizing provenance, collection conditions, and preprocessing steps. In this section, we outline several data requirements to make similar pavement overhead image data AI-ready. The *MoPac+* AI-Ready Datasheet is envisioned as a structured methodology for curating and processing pavement imagery that explicitly addresses the challenges of training robust DNNs for real-world infrastructure monitoring. This approach bridges critical gaps between raw data acquisition and model-ready inputs through three key innovations: (1) sensor-optimized 3D/2D fusion, (2) adaptive preprocessing for distress feature enhancement, and (3) metadata-driven class balancing.

4.1. Data Acquisition and Annotation

The data acquisition system employs a specialized 3D laser camera that captures simultaneous 2D (Intensity) and 3D (Range) values via a synchronized line-scan camera. This configuration adheres to the AASHTO 2D/3D pavement format standards [42], ensuring consistency across segments spanning the full lane width (approximately 4 meters) and a longitudinal range of 2 to 12 meters, depending on driving speed. The triangulation-based depth calculation can produce temporally aligned 2D/3D pairs that preserve the spatial relationships critical for crack-geometry analysis. After scanning a road segment, the system converts the raw data into 2D and 3D grayscale images. It stores them in an industry-standard format, with both image types sharing the same resolution and spatial coordinates within a single file of 1536 x 900 pixels (width x height). The inclusion of 3D height maps offers distinct advantages over conventional 2D approaches, as illustrated in Figure 4 with a sample 2D and corresponding 3D image of the same road segment.

The critical aspect of making the data AI-ready is defining labels that are both semantically meaningful to domain experts and compatible with computer vision algorithms. This dual focus ensures that the labeled data reflects expert knowledge and facilitates efficient training and accurate inference by deep learning models. After data acquisition, the 2D and 3D images were annotated with standard JCP

distress classifications using in-house software developed by Dr. Gong. During this process, the annotators were provided with 2D and 3D imagery, as well as Google Street View imagery, of specific pavement locations. They were then tasked with annotating bounding boxes, delineated by distress classification, over the entire extent of the distress classification —including the background —as tightly as possible. During this stage, precise, consistent, and unbiased annotation is paramount for the success of the DNN models. The software then converts these annotations into the YOLO annotation format [43] for storage, modification, and subsequent use as input to an AI model. This study selects 20 highway sections with Joint Concrete Pavement (JCP) surfaces, totaling 7,750 pairs of 2D and 3D images, containing 16,943 annotations (or "instances"), and is titled *JCPv3* (Figure 6). *JCPv3* contains 7,079 image pairs with annotated distress and 671 pairs with no distress present, and all of the images are the same standard dimension of 1536 x 900 pixels (w x h). After initial experiments and evaluations from studies such as *MoPac*, it was determined that more robust classes are needed and that some classes are too closely matched to others. For these reasons, the significant differences between *JCPv1* and *JCPv3* were that: Spall was removed, distributed across other classes; S-Trans was added; Joint was divided into Joint and S-Edge for visual similarity.

4.2. Data Preparation

The *JCPv3* dataset consists of 14 distinct classes of distress, but, due to the limitation of Deep Learning requiring a minimum amount of examples to perform predictably, the distress classes D-Cracking (5 annotations) and Popout (39 annotations) were removed from this implementation. Table 1 illustrates how we split the distress classes into *Recurring* (over 500 instances) and *Rare* (under 500 instances) categories, with representative 2D intensity images and corresponding 3D range images provided for each class. This class splitting reveals the critical challenges inherent to pavement distress detection: (i) Recurring classes (e.g., Slab Edge) dominate the distribution, while rare classes (e.g., Punchout) exhibit severe under-representation ($\leq$ 8% of total instances); (ii) Longitudinal cracks and joint features demonstrate $\leq$ 12% visual similarity in gradient histograms, confounding traditional detection approaches.

Dataset Stratification The dataset was partitioned into training and testing subsets, ensuring that at least 20 instances per class were present in the test set. This threshold was determined based on the dataset's statistical distribution, thereby supporting reliable and meaningful model evaluation. Algorithm 1 describes our reproducible stratified train-and-test splitting strategy with per-class balancing. We first parse all annotation files using `load_class_instances` to build a mapping $\mathcal{M} = \{c \mapsto \mathcal{F}_c\}$, where $\mathcal{F}_c$ is the list of image filenames that contain at least one instance of class c. For each class c, we randomly shuffle $\mathcal{F}_c$ using a fixed seed to ensure determinism. We then determine the number of test images to draw from that class by combining (i) an absolute minimum requirement and (ii) a proportional requirement based on the desired test ratio. Concretely, let $N_c = |\mathcal{F}_c|$ and let r be `test_ratio`. We compute

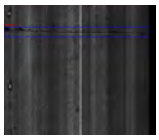
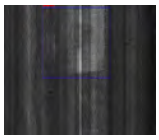
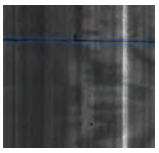
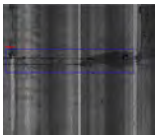
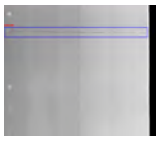
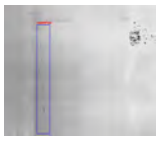
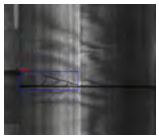
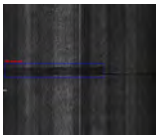
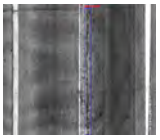
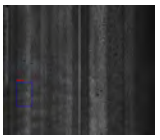
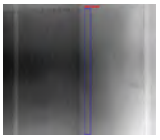
$$t_c = \min\Big(N_c,\ \max\big(\texttt{min_per_class},\ \lceil r\, N_c \rceil\big)\Big),$$

i.e., for each class we select the *greater* of `min_per_class` and the ratio-based target, but we cap the result at N_c so we never request more test samples than the class provides. The first t_c shuffled filenames from $\mathcal{F}_c$ are added to a global set T (a set union), ensuring that an image is included in the test split if it is selected by *any* class it contains; duplicates are automatically removed. Finally, we iterate over all images and assign an image to the test split if its filename is in T; otherwise, it is assigned to the training split, copying the corresponding image and label files to the respective directories. After applying Algorithm 1, our training split contains 6052 images, and our test split contains 2358 images. This approach ensures a balanced representation of both common and rare classes in the test set, facilitating

Algorithm 1: Stratified dataset splitter with per-class balancing

```
Don't Print Semicolon   Function stratified_split(class_to_files, min_per_class, test_ratio):
    Initialize empty set: test_files;
    foreach class, files in class_to_files do
        Shuffle file list;
        num_test ← min(length, max(min_per_class, test_ratio * total));
        Add first num_test files to test_files;
    return test_files;
Function split_dataset(image_dir, annotation_dir, output_root, test_ratio):
    Set random seed to ensure reproducibility;
    Create 'train/images', 'train/labels', 'test/images', and 'test/labels' directories under output_root;
    class_to_files ← load_class_instances(annotation_dir);
    test_files ← stratified_split(class_to_files, min_per_class=20, test_ratio);
    foreach img_file in image_dir do
        if file does not end with '.png' then
            continue
        base ← filename without extension;
        ann_file ← base + '.txt';
        if img_file is in test_files then
            split ← 'test'
        else
            split ← 'train'
        Copy img_file to split/images;
        Copy ann_file to split/labels;
```

Table 1
Displays examples of the Recurring classes (top row) and Rare classes (bottom row) within the JCP dataset, providing both Intensity (2D) and Range (3D) imagery along with their annotations (blue bounding boxes). Notably, the same annotations are used in both the 2D and 3D images.

JCP Recurring Classes: Number of labeled instances is enough for the robust modeling						
Label →	Joint	L-Crack	T-Crack	C-Patch	S-Edge	F-Joint
Instances (Orig.)	4,127	1,097	649	875	8357	519
Instances (After Aug.)	4,527	1,165	668	1562	9063	624
Description	Joint Crack	Longitudinal Crack	Transversal Crack	Concrete Patch	Slab Edge	Failed Joint
Intensity						
Range						
JCP Rare Classes: Number of labeled instances is not enough for robust modeling						
Label →	C-Break	A-Patch	S-Trans	FC-Patch	S-Long	P-Out
Instances (Orig.)	300	346	50	42	441	96
Instances (After Aug.)	300	400	85	72	617	192
Description	Corner Break	Asphalt Patch	Sealed Transverse	Failed Concrete Patch	Sealed Longitudinal	Punchout
Intensity						
Range						

standardized benchmark comparisons and enhancing the dataset's suitability for AI-driven analysis. As the dataset exhibits a long-tailed distribution with several rare classes, we do not create a fixed validation set. Carving out a fixed validation set from the training data would further reduce supervision for these rare categories, potentially degrading detection performance and stability.

Image Augmentation The augmentation increases the sample size and makes the crack lines thicker and more visible (Note: We perform augmentations only for rare classes with fewer than 200 instances). We perform a series of augmentations on the images. However, we rejected any morphological augmentation that could significantly alter the shape or size of the cracks. The augmentations used in the preprocessing are as follows: 1) Horizontal Flip, 2) Gaussian Noise, 3) CLAHE, 4) CANNY, 5) Random Brightness Contrast, and 6) Elastic Transform. We selected four random augmentations from these six types for preprocessing. This augmentation process improved overall model performance on the test dataset; the results are presented in the ablation study in Section 5.

5. Experiments

In this section, we present a comprehensive evaluation of our proposed *MoPac+* model against state-of-the-art approaches using our experimental dataset. Our dataset comprises 12 distinct classes, meticulously preprocessed through the augmentation pipeline detailed in Section 4. To ensure rigorous evaluation, we employ an instance-aware train-test splitting strategy as formalized in Algorithm 1. Additionally, we maintained consistent values for common hyperparameters, including learning rate, number of epochs, and IOU, across all comparison methods. For validation, we use Holdout validation to assess a model's performance on unseen data by early stopping to prevent overfitting, hyperparameter tuning, and model checkpoint selection based on validation mAP. Specifically, we reserve 15% of the training dataset as the holdout set maintaing classwise stratification.

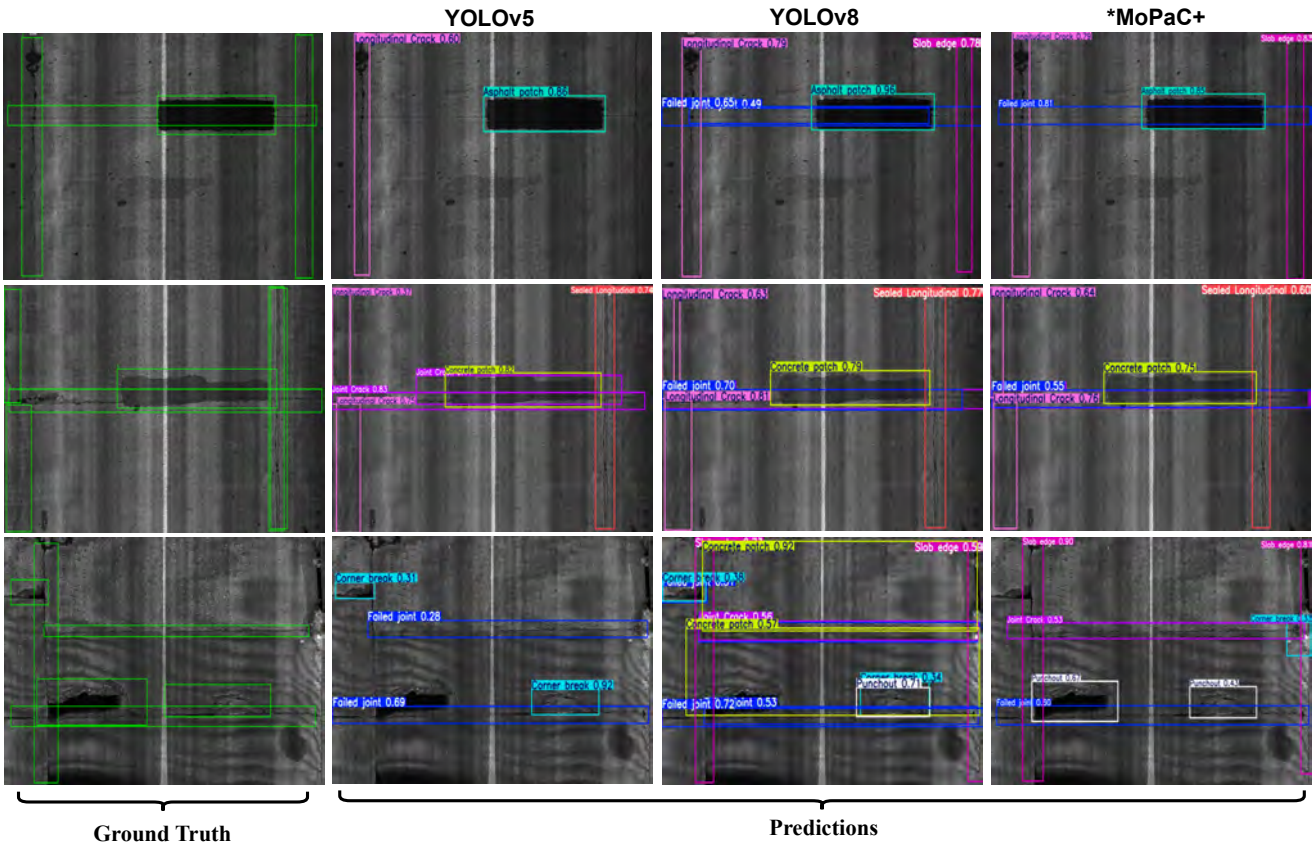


Figure 7: SOTA detection comparison with proposed **MoPac+* model. Depicting model prediction done on a few randomly selected test samples (rows), comparing the ground truth, YOLOv5, YOLOv8, and *MoPac+, respectively (columns).

5.1. Setup

The *MoPac+* architecture is implemented in Python, extending the official YOLOv8 implementation from ULTRALYTICS within the PyTorch deep learning framework. All experiments are conducted on hardware equipped with an NVIDIA Titan XP 1080 GPU (12GB VRAM), an Intel Core i9-11900K processor (3.50 GHz, 16 cores), and 167GB of DDR4 system memory. We maintain consistent training parameters as presented in the Table 2 across all experiments. Our analysis employs five principal metrics to assess detection performance:

$$\text{mAP} = \frac{1}{C}\sum_{c=1}^{C} \text{AP}_c, \qquad \text{AP}_c = \int_0^1 p_c(r)\,dr \tag{6}$$

- **Recall (R)** - Ratio of correctly identified positive instances to all actual positives.
- **Average Recall (AR)** - Recall averaged across multiple Intersection-over-Union (IoU) thresholds.
- **Precision (P)**: Proportion of correct detections among all positive predictions.
- **Average Precision (AP)** - Area under the Precision-Recall curve per class, as defined in Equation 6.

Table 2
Important hyperparameter configurations used across all experiments.

Category	Setting
Epochs	25
Input Size	640
Batch Size	16
Optimizer	AdamW
Momentum	0.937
Weight Decay	0.0005
Learning Rate	0.0001
Regression Loss Weight (W_r)	0.1
Classification Loss Weight (W_c)	0.5
Class Loss Gain	0.5
Box Loss Gain	0.5

- **Mean Average Precision (mAP)** - Mean AP across all classes, computed at a certain IoU threshold, as defined in Equation 6.

The mAP metric, defined in Equation 6, serves as our primary benchmark for model comparison, with class-specific

Table 3
Table for the average performance of different SOTA Models for all 12 classes. **Note: * denotes multimodal training with Intensity+Range Imagery. Also, the FPS is reported using the NVIDIA RTX 6000 Ada GPU for end-to-end inference.**

Anchor	Algorithm	Prec.	Recall	mAP 50	mAP 50-95	L. Param (M.)	GFLOPS	FPS
Yes	YOLOv5m	0.684	0.541	0.606	0.327	**21.22**	**49.50**	280
Yes	YOLOv7	0.672	0.542	0.551	0.286	36.97	104.70	212
No	CenterNet2	0.697	0.585	0.644	0.320	58.10	210.20	40
No	DEFORM. DETR	0.683	0.551	0.627	0.311	42.48	173.63	22
No	YOLOv8m (Baseline)	0.675	0.594	0.625	0.345	25.85	78.70	244
No	*MoPac+*(Intensity)	0.713	0.672	0.693	0.381	26.11	79.88	232
No	**MoPac+*(I+R)	**0.716**	**0.730**	**0.726**	**0.387**	26.11	79.88	232

AP scores providing detailed insights into detection capabilities across both common and rare categories. To address class imbalance, we stratify our evaluation into two distinct groups: (i) *Frequent Classes*: Commonly occurring categories with abundant training samples, and (ii) *Rare Classes*: Infrequently observed categories requiring specialized analytical consideration. This stratification enables separate analysis of model performance for high-frequency classes ($N_c \geq \tau$) and low-frequency classes ($N_c < \tau$), where N_c denotes the number of instances for class c and τ is the frequency threshold.

5.2. MoPac+ vs. SOTA Performance Comparison

Table 3 presents a comparative evaluation of state-of-the-art (SOTA) object detection models, alongside the proposed *MoPac+* framework. It is important to note that YOLOv5 and YOLOv7 [5, 44] utilize anchor-based detection mechanisms, whereas CenterNet2, DETR, YOLOv8m (serving as the baseline), and *MoPac+* operate without anchors. This study combines anchor-based and anchor-free approaches to rigorously assess the efficacy of anchor-free detectors in accurately identifying objects with diverse shapes. We also present Multimodal performance, along with Intensity-Only MoPac+ performance, to demonstrate the limitations of single-modality training. From the results summarized in Table 3, YOLOv5 achieves a competitive Precision score of 0.68 relative to YOLOv7 and YOLOv8. The CenterNet2 becomes the closest competitor with an overall AP of 0.69, and an mAP(IOU=0.5) of 0.64. The transformer-based DETR performs very close to CenterNet2 in terms of Prec., however, the Recall drops by $\Delta 3\%$. The proposed **MoPac+* model attains the highest Precision, surpassing the nearest SOTA (CenterNet2) by 2%. MoPac+ demonstrates the benefit of multimodal training, achieving higher Recall and mean Average Precision (mAP) at IoU = 0.50. Specifically, *MoPac+ attains 73% Recall and 72.6% mAP, improving over the single-modality baseline by 6% and 3.3%, respectively. These gains suggest that fusing complementary modalities reduces missed detections that arise when a single modality provides insufficient or ambiguous visual cues. Moreover, as a state-of-the-art candidate, **MoPac+* outperforms the strongest competing method, CenterNet2, by 15% in Recall and 8% in mAP. Notably, anchor-free methods achieve higher recall and significantly outperform anchor-based detectors in mAP. Additionally, the observed mAP@0.50 exceeds the corresponding mAP averaged over IoU thresholds 0.50 to 0.95, consistent with prior findings reported in [45] for overhead imagery. Figure 7 illustrates the qualitative result comparison for *MoPac+ with relevant SOTA models. Figure 7 (fourth row) illustrates two instances of punchout distress, which are visually distinct from a computer vision perspective. This diversity poses a classification challenge for most methods. Importantly, the proposed **MoPac+* model correctly detects both instances, attributable to its robust feature-extraction backbone, extensive data augmentation during training, and the integration of hard-example mining techniques. In Figure 8, we plot the classification and bbox train loss curve for the baseline and our proposed multimodal *MoPac+ model. It is evident that multimodal training yields greater stability than the baseline model. Also, the monotonic decrease in the loss suggests improved regularization that prevents overfitting on the test set. Overall, our proposed model achieves a balanced performance gain across distress sizes and recurrences in the dataset.

As one of our primary motivations was to propose a model that is lightweight enough to deploy on resource-constrained devices, in Table 3 we present the L. Params, inference rate (FPS), and GFLOPS to compare against SOTA models. From Table 3, we see that YoloV5 requires the least computational resources. On the other hand, despite its competitive performance, CenterNet2 requires the most computational resources. The inference rate (FPS) for CenterNet2 is 40 FPS; by contrast, YOLOv8m achieves 244 FPS under similar settings, which is a standard real-time response for common computer vision applications. The proposed MoPac+ also demonstrates a standard real-time response, achieving 232 FPS for end-to-end inference. Hence, we chose YOLOv8m as the baseline model because it achieves satisfactory performance for road crack detection and requires significantly less computational power.

5.3. Class-wise Performance Comparison

In this section, we conduct the class-wise analysis based on the *Recurring* and *Rare* distress category classifications

Table 4
Performance of the SOTA models per recurring label (top) and rare label (bottom) on the test set. Here, AP and AR were used as the evaluation metrics. **Note: * denotes multimodal training with Intensity+Range Imagery.**

Recurrent Label	J-Crack		L-Crack		T-Crack		C-Patch		S-Edge		F-Joint	
Method ↓	AP	AR	AP	AR	AP	AR	AP	AR	AP	AR	AP	AR
YOLOv5m	0.85	0.86	0.64	0.65	**0.74**	0.55	0.72	0.83	0.84	0.90	0.61	0.67
CenterNet2	0.71	0.42	0.69	**0.85**	0.66	0.63	0.77	0.80	0.80	0.91	0.61	0.50
DEFORM. DETR	0.73	0.80	0.66	0.78	0.67	**0.68**	0.70	0.81	0.84	0.89	0.58	0.47
YOLOv7	0.69	0.18	0.62	0.76	0.70	0.64	0.67	**0.86**	0.83	**0.95**	0.60	0.56
YOLOv8s	0.79	0.87	0.60	0.68	0.67	0.57	0.70	0.81	0.89	0.93	0.49	0.73
YOLOv8m (Baseline)	0.80	0.87	0.60	0.73	0.67	0.60	0.74	0.84	0.91	0.93	0.49	0.75
MoPac+ (Intensity)	**0.94**	0.73	0.73	0.67	0.64	0.58	0.83	0.77	**0.97**	0.80	0.62	**0.75**
**MoPac+* (I+R)	0.91	**0.92**	**0.81**	0.78	0.72	0.65	**0.87**	0.81	0.93	0.94	**0.64**	0.73
Rare Label	C-Break		A-Patch		S-Trans		FC-Patch		S-Long		P-Out	
Method ↓	AP	AR	AP	AR	AP	AR	AP	AR	AP	AR	AP	AR
YOLOv5m	0.60	0.32	0.64	0.54	0.95	0.20	0.90	0.18	0.73	0.82	0.50	0.47
CenterNet2	0.55	0.40	0.70	0.56	0.88	0.30	0.97	0.26	0.74	0.85	**0.64**	0.63
DEFORM. DETR	**0.62**	0.42	0.60	0.55	0.75	0.25	0.90	0.30	0.71	0.86	0.58	0.59
YOLOv7	0.58	0.41	0.47	0.57	**0.98**	0.18	**0.98**	0.18	0.62	0.87	0.24	**0.72**
YOLOv8s	0.48	0.33	0.72	0.56	0.12	0.22	0.44	0.18	0.74	0.80	0.35	0.28
YOLOv8m (Baseline)	0.50	0.33	0.74	0.56	0.12	0.25	0.44	0.16	0.76	0.80	0.38	0.36
MoPac+ (Intensity)	0.50	**0.49**	0.62	**0.66**	0.45	0.70	0.90	0.30	0.83	0.90	0.36	0.40
**MoPac+* (I+R)	0.60	0.47	**0.78**	**0.66**	0.80	**0.87**	0.84	**0.30**	**0.84**	**0.90**	0.60	0.46

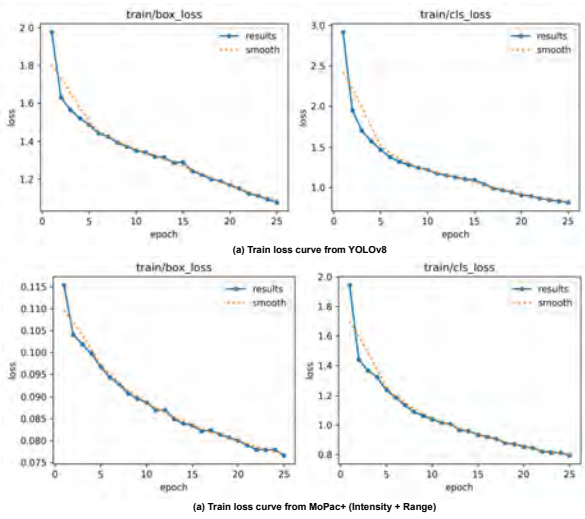


Figure 8: The classification and bbox train loss curve for the baseline and our proposed multimodal MoPac+ model. The loss curve shows superior stability while training with our proposed model. Also, the monotonic decrease in the loss suggests improved regularization that prevents overfitting on the test set.

as defined previously in Section 4 and Table 1. Using Average Precision (AP) and Average Recall (AR), Table 4 shows the performance of *Recurring* and *Rare* classes with SOTA and proposed *MoPac+* methods.

Recurring Classes: The dominant performance comes from our proposed *MoPac+* model (See Table 4), with YOLOv7 being the closest competitor in detecting the recurrent courses. YOLOv5 demonstrates superior performance in T-Crack detection, achieving the highest AP of 0.74 and exhibiting the best precision for F-Joint detection among SOTA models at 0.61. However, its AR performance varies considerably across crack types. On the other hand, the lightweight DEFORM. DETR achieves the competitive AP and the highest AR (0.85) for the L-Crack class. YOLOv7 performs well in terms of Recall, achieving the highest AR scores for C-Patch (0.86) and S-Edge (0.95) detection. YOLOv7's strength lies in identifying common instances of distress types, albeit at the expense of lower precision. **MoPac+* achieves the highest AP scores across four out of six crack types: J-Crack (0.91), L-Crack (0.84), C-patch (0.87), and F-Joint (0.64). For distress types such as S-Edge, C-Patch, and J-Crack, we achieved notable precision with our **MoPac+* model. The Recall for *MoPac also shows impressive results across several classes, narrowing the gap to SOTA performance. Thus, even in a detailed evaluation of recurrent distress types presented in Table 4, our method outperforms the available SOTA methods for our dataset.

Rare Classes: The findings from Table 4 were interesting and closely aligned with our hypothesis regarding the use of multimodal learning and hard-example mining for these classes. For Rare classes, the shapes can be challenging, and the annotation instance count was minimal. YOLOv5 demonstrates strong Precision capabilities, achieving the highest AP for C-Break detection (0.60) and competitive performance on A-Patch (0.64), S-Trans (0.95), and FC-Patch (0.90). YOLOv7 shows exceptional precision for S-Trans (0.98) and FC-Patch (0.98), achieving the highest AP among all methods. Despite this, YOLOv7 did not perform well on these classes, as reflected in its AR scores on S-Trans (0.18) and FC-Patch (0.18), indicating limitations when

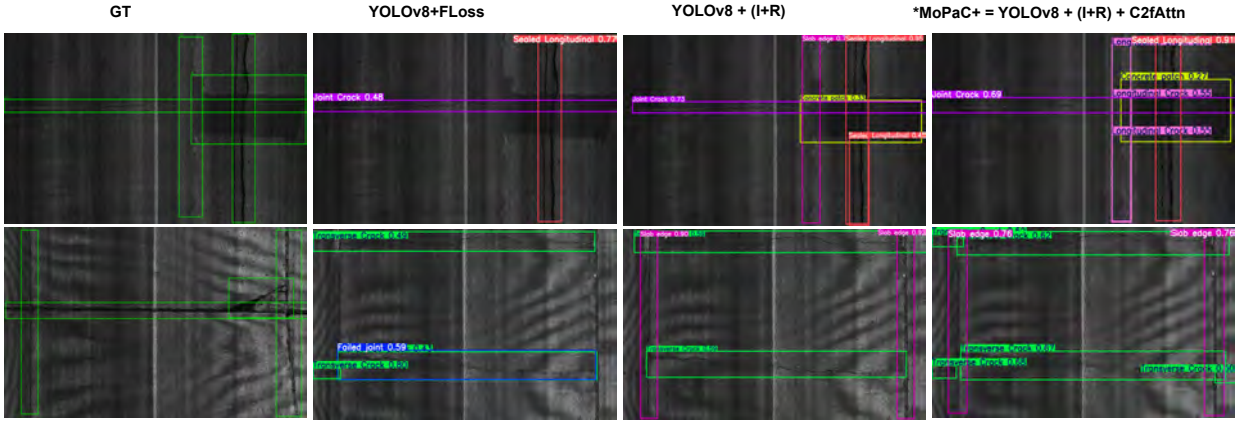


Figure 9: Ablation Study for different modules in the proposed *MoPac+* pipeline. We compare the detection results with those of the baseline YOLOv8 architecture.

Table 5

Ablation study for different modules of our *MoPac+* method where Baseline (YOLOv8m), Data Augmentation (D Aug.), Focal Loss (F. Loss) is used for hard-example mining, Weight Balance (Weight B) is the balance between classes. and reg. loss, and C2fAttn is for context-aware feature aggregation. **Note: * denotes multimodal training with Intensity+Range Imagery.**

Model (IOU=0.5)	Data Aug.	Weight B	F. Loss	C2fAttn	Prec.	Recall	mAP 50	mAP 50-95
Baseline (Intensity)					0.668	0.592	0.621	0.335
w Aug (Intensity)	✓				0.657	0.601	0.635	0.345
w Aug (I+R)	✓				0.663	0.634	0.647	0.349
w Weight B. (I+R)	✓	✓			0.680	0.643	0.675	0.376
w F.Loss (I+R)	✓		✓		0.675	0.638	0.664	0.356
w C2fAttn (I+R)	✓			✓	0.679	0.677	0.674	0.365
*MoPac+ $IOU = 0.3$	✓	✓	✓	✓	0.708	0.722	0.718	0.375
*MoPac+ $IOU = 0.5$	✓	✓	✓	✓	**0.716**	**0.730**	**0.726**	**0.387**
*MoPac+ $IOU = 0.7$	✓	✓	✓	✓	0.695	0.684	0.690	0.371

trained with a small sample size. CenterNet2 [46] is a point-based network that is efficient for pixel-level feature extraction. Hence, it shows relatively better performance on small classes, such as FC-Patch and P-Out, achieving ARs of 0.85 and 0.64, respectively, compared with DEFORM. DETR also demonstrates strong performance on small classes and remains the closest competitor to the MoPac+ method. The proposed *MoPac+* method outperformed competitors across five of six crack types, with a substantial margin. It achieved the highest Recall across multiple categories: C-Break (0.49), A-Patch (0.66), S-Trans (0.87), FC-Patch (0.30), and S-Long (0.90), aided by complementary range-imaging supervision.

5.4. Ablation Study

This section presents an ablation study to assess the effects of the enhancements in our proposed method on DNN performance. Using YOLOv8 as our baseline, we isolate the improvements of the *MoPac+* pipeline across different modules and explain the rationale for the method by analyzing its performance and impact on the overall modeling process. We chose YOLOv8 with a CSPDarknet53 backbone because of its superior performance for a range of shapes and better performance for small objects. The baseline model achieves an mAP of 0.621 and a recall of 0.592, providing a good starting point. To achieve balanced performance across recurrent and rare classes and to maximize the benefits of the proposed modules, we also perform train-time data augmentation. The YOLO training pipeline applies augmentations stochastically during each epoch, randomly selecting subsets from the augmentation pool (Horizontal Flip, Gaussian Noise, Blurring Filter, Random Brightness/Contrast, Elastic Transform). Table 5 shows that preprocessing improves Recall and mAP by at least 1.2%. Next, to detect more missed objects during intensity-only training, we perform multimodal training with I+R, which increases Recall by 3%. The precision and mAP also improve by +1% as shown in Table 5. In Figure 9, we plot the effectiveness of the multimodal training technique with YOLOv8. From this figure, we see that the multimodal training detects more objects with incomplete information and ambiguous properties. The shapes of distress can be clustered into very few preprocessing steps, so the regression task is less complex in our applications. We aim to identify the types of distress that are challenging, as discussed in

Table 6
Ablation study on the classification and regression loss weight. We perform a selective search across both parameters using a predefined set of values.

Reg. W. (W_r)	Cls. W. (W_c)	Prec.	Recall	mAP 50
0.1	0.5	0.716	0.730	0.726
0.3	0.5	0.678	0.705	0.662
0.7	0.5	0.584	0.540	0.563
0.5	0.7	0.668	0.615	0.633

earlier sections. To resolve this issue, we perform weight balancing on the regression and classification pipeline. Table 6 presents ablation studies on the classification and regression weight-balancing parameters. We perform ablation studies with a few values of W_r and W_c ranging from [0.1 to 0.7] to select the optimal values. From Table 6, it is evident that the $W_r = 0.1$ and $W_c = 0.5$ achieve optimal performance for the experimental dataset. With the selected values, we achieved an improvement of more than +2% across all evaluation matrices.

To address the challenges of rare-class detection, we implemented a class-balanced focal loss function that dynamically penalizes misclassifications according to the frequency of class instances in the training dataset. Our hard-example mining strategy yielded significant improvements, achieving +2% AP and +3% recall gain for rare classes compared to baseline approaches. Figure 9 provides qualitative visualizations of this improvement through comprehensive ablation studies. Finally, we introduce the C2fAttn module to improve performance where cracks are often *thin*, *low-contrast*, and easily confused with sealed joints, stains, or surface texture. Introducing C2fAttn gains approx. 5% in Recall and +4% in mAP across all classes. To optimize post-processing, we empirically evaluated Non-Maximum Suppression (NMS) thresholds at 0.3, 0.5, 0.7 IOU thresholds. Our analysis revealed very similar performance across thresholds from 0.3 to 0.5 ($\Delta mAP < 1\%$), with an IOU of 0.5 emerging as optimal for distress detection. Using this configuration, our multimodal *MoPac+* framework achieved state-of-the-art performance with 0.726 mAP and 0.730 recall, establishing a new SOTA benchmark for imbalanced pavement distress detection tasks.

6. Conclusion and Future Work

In this paper, we propose **MoPac+*, a state-of-the-art crack localization and identification framework that effectively recognizes different types of cracks in the pavement from intensity and range images. We have prepared an AI-ready multimodal dataset for a detection benchmark across six recurrent and six rare distress types. First, we have improved the labeling of our training dataset and defined categories that are visually meaningful for both human experts and the automated system. We introduce an anchor-free multimodal YOLOv8 pipeline that robustly detects joints and asphalt patches across larger road patches. We have employed context-guided feature aggregation to improve performance on thin and low-contrast classes, and the weight-balance technique to place greater emphasis on the task-specific detection pipeline. Next, we propose hard-example mining with focal loss to improve performance on rare classes. Using an IOU threshold of 0.5, we achieve the highest Recall of **0.730** and the highest mean average precision of **0.726**, outperforming the latest state-of-the-art (SOTA) methods. The next step for *MoPac+ is to continue collecting additional multimodal pavement data from different geographic regions and pavement types to enable more comprehensive generalization studies. Also, we plan to improve detection performance by applying pixel-wise feature enhancement, integrating a super-resolution technique, and channel-wise self-attention to the early-fusion input features.

AUTHOR CONTRIBUTIONS

Debojyoti led the machine learning implementation, data augmentation, and experiments, and contributed to manuscript preparation. Scouten created tables and figures and contributed to manuscript preparation and editing. Gong led the dataset's creation and manual annotation, produced figures, and contributed to the literature review, manuscript preparation, and editing. Tešić designed the experiments and data analysis, and contributed to the literature review, the manuscript preparation, and editing. Wang conceived the research idea, supervised the project, and contributed to the manuscript's writing and editing.

Acknowledgments

The study has been partially supported by funding from the following sources: TxDOT Project #0-7150 and NSF Project 2213694.

References

[1] T. Texas Department of Transportation, 2023 roadway inventory report summary mileage, traffic and environmental statistics (2023). URL https://www.txdot.gov/content/dam/docs/tpp/roadway-inventory/road-inventory-annual-report-2023.pdf

[2] X. Chen, C. Liu, L. Chen, X. Zhu, Y. Zhang, C. Wang, A pavement crack detection and evaluation framework for a uav inspection system based on deep learning, Applied Sciences 14 (3) (2024) 1157.

[3] A. Saberironaghi, J. Ren, Depthcracknet: A deep learning model for automatic pavement crack detection, Journal of Imaging 10 (5) (2024) 100.

[4] S. Zhou, D. Yang, Z. Zhang, J. Zhang, F. Qu, P. Punetha, W. Li, N. Li, Enhancing autonomous pavement crack detection: Optimizing yolov5s algorithm with advanced deep learning techniques, Measurement 240 (2025) 115603.

[5] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: Unified, real-time object detection, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 779–788.

[6] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, S. Zagoruyko, End-to-end object detection with transformers, in:

European conference on computer vision, Springer, 2020, pp. 213–229.

[7] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: Towards real-time object detection with region proposal networks, Advances in neural information processing systems 28 (2015).

[8] E. El-Din Hemdan, M. Al-Atroush, A review study of intelligent road crack detection: Algorithms and systems, International Journal of Pavement Research and Technology (2025) 1–31.

[9] T. Rahman, T. Mahmud, S. Roy, M. M. Setara, M. S. Hossain, K. Andersson, Enhancing road safety: A deep learning approach to automated crack detection, in: 2024 Second International Conference on Inventive Computing and Informatics (ICICI), IEEE, 2024, pp. 405–411.

[10] L. Zheng, J. Xiao, Y. Wang, W. Wu, Z. Chen, D. Yuan, W. Jiang, Deep learning-based intelligent detection of pavement distress, Automation in Construction 168 (2024) 105772.

[11] L. Deng, H. Yuan, L. Long, P.-j. Chun, W. Chen, H. Chu, Cascade refinement extraction network with active boundary loss for segmentation of concrete cracks from high-resolution images, Automation in Construction 162 (2024) 105410.

[12] H. Gong, J. Tešić, J. Tao, X. H. Luo, F. Wang, Automated pavement crack detection with deep learning methods: What are the main factors and how to improve the performance?, Transportation Research Record 2677 (10) (2023) 311–323. doi:10.1177/03611981231161358.

[13] F. Liu, G. Xu, Y. Yang, X. Niu, Y. Pan, Novel approach to pavement cracking automatic detection based on segment extending, in: 2008 International Symposium on Knowledge Acquisition and Modeling, IEEE, 2008, pp. 610–614.

[14] M. Gavilán, D. Balcones, O. Marcos, D. F. Llorca, M. A. Sotelo, I. Parra, M. Ocaña, P. Aliseda, P. Yarza, A. Amírola, Adaptive road crack detection system by pavement classification, Sensors 11 (10) (2011) 9628–9657.

[15] Q. Li, X. Liu, Novel approach to pavement image segmentation based on neighboring difference histogram method, in: 2008 congress on image and signal processing, Vol. 2, IEEE, 2008, pp. 792–796.

[16] N. Hu, J. Yang, X. Jin, X. Fan, Few-shot crack detection based on image processing and improved yolov5, Journal of Civil Structural Health Monitoring 13 (1) (2023) 165–180.

[17] G. Yu, X. Zhou, An improved yolov5 crack detection method combined with a bottleneck transformer, Mathematics 11 (10) (2023) 2377.

[18] A. M. Roy, J. Bhaduri, Densesph-yolov5: An automated damage detection model based on densenet and swin-transformer prediction head-enabled yolov5 with attention mechanism, Advanced Engineering Informatics 56 (2023) 102007.

[19] P. Wu, J. Wu, L. Xie, Pavement distress detection based on improved feature fusion network, Measurement 236 (2024) 115119.

[20] G. Ye, J. Qu, J. Tao, W. Dai, Y. Mao, Q. Jin, Autonomous surface crack identification of concrete structures based on the yolov7 algorithm, Journal of Building Engineering 73 (2023) 106688.

[21] V. Pham, D. Nguyen, C. Donan, Road damage detection and classification with yolov7, in: 2022 IEEE International Conference on Big Data (Big Data), IEEE, 2022, pp. 6416–6423.

[22] G. Hu, B. Hu, Z. Yang, L. Huang, P. Li, Pavement crack detection method based on deep learning models, Wireless Communications and Mobile Computing 2021 (1) (2021). doi:10.1155/2021/5573590.

[23] X. Cao, P. Yuan, B. Feng, K. Niu, Cf-detr: Coarse-to-fine transformers for end-to-end object detection, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 36, 2022, pp. 185–193.

[24] F. Guo, J. Liu, Q. Xie, H. Yu, A two-stage framework for pixel-level pavement surface crack detection, Engineering Applications of Artificial Intelligence 133 (2024) 108312.

[25] C. Han, H. Yang, T. Ma, S. Wang, C. Zhao, Y. Yang, Crackdiffusion: A two-stage semantic segmentation framework for pavement crack combining unsupervised and supervised processes, Automation in Construction 160 (2024) 105332.

[26] S. D. Son Dong Nguyen, T. S. Tran, V. P. Tran, H. J. Lee, M. J. Piran, V. P. Le, Deep learning-based crack detection: A survey, International Journal of Pavement Research and Technology 16 (2022) 943–967. doi:10.1007/s42947-022-00172-z.

[27] D. Li, Q. Xie, X. Gong, Z. Yu, J. Xu, Y. Sun, J. Wang, Automatic defect detection of metro tunnel surfaces using a vision-based inspection system, Advanced Engineering Informatics 47 (2021) 101206.

[28] J. Huyan, W. Li, S. Tighe, J. Zhai, Z. Xu, Y. Chen, Detection of sealed and unsealed cracks with complex backgrounds using deep convolutional neural network, Automation in Construction 107 (2019) 102946.

[29] D. H. Kang, Y.-J. Cha, Efficient attention-based deep encoder and decoder for automatic crack segmentation, Structural Health Monitoring 21 (5) (2022) 2190–2205.

[30] Z. Liu, J. Wang, X. Teng, N. Li, Fine-tuning large vision model for multimodal fusion in asphalt pavement crack segmentation, International Journal of Pavement Engineering 26 (1) (2025) 2526158.

[31] L. Zhao, T. Xu, Multimodal perception-guided intelligent pavement crack detection approach: a novel combining vision and semantic supervision framework, Measurement (2025) 119333.

[32] J. Zhai, Z. Sun, J. Huyan, H. Yang, W. Li, Automatic pavement crack detection using multimodal features fusion deep neural network, International Journal of Pavement Engineering 24 (2) (2023) 2086692.

[33] A. Scouten, H. Gong, J. Tešić, F. Wang, Deep learning pipeline for modeling pavement cracks with an imbalanced data set (mopac), Poster presentation at the Transportation Research Board (TRB) 104th Annual Meeting, presentation Number: TRBAM-25-06415 (2025).
URL https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22848

[34] G. Jocher, the Ultralytics team, YOLOv8: Ultralytics Official Implementation, https://github.com/ultralytics/ultralytics, accessed: 2025-06-07 (2023).

[35] A. Bochkovskiy, C.-Y. Wang, H.-Y. M. Liao, Yolov4: Optimal speed and accuracy of object detection, arXiv preprint arXiv:2004.10934 (2020).

[36] D. Biswas, J. Tešić, Small object difficulty (sod) modeling for objects detection in satellite images, in: 2022 14th International Conference on Computational Intelligence and Communication Networks (CICN), IEEE, 2022, pp. 125–130.

[37] D. Biswas, J. Tešić, Binarydnet53: a lightweight binarized cnn for monkeypox virus image classification, Signal, Image and Video Processing 18 (10) (2024) 7107–7118.

[38] M. Yaseen, What is YOLOv8: An in-depth exploration of the internal features of the next-generation object detector, arXiv cs.CV (Aug. 2024). arXiv:2408.15857.
URL https://arxiv.org/abs/2408.15857

[39] J. Redmon, A. Farhadi, Yolov3: An incremental improvement, arXiv preprint arXiv:1804.02767 (2018).

[40] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE international conference on computer vision, 2017, pp. 2980–2988.

[41] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, K. Crawford, Datasheets for datasets, Commun. ACM 64 (12) (2021) 86–92. doi:10.1145/3458723.
URL https://doi.org/10.1145/3458723

[42] AASHTO, Standard specification for file format of two-dimensional and three-dimensional (2d/3d) pavement image data (2023).

[43] G. Jocher, A. Chaurasia, J. Qiu, Ultralytics YOLO (Jan. 2023).
URL https://github.com/ultralytics/ultralytics

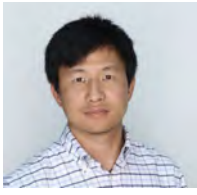
[44] J. Redmon, A. Farhadi, Yolo9000: better, faster, stronger, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 7263–7271.

[45] N. Warren, B. Garrard, S. E., Tešić, J., Transfer learning of deep neural networks for visual collaborative maritime asset identification, in: 2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC), 2018, pp. 246–255.

[46] X. Zhou, V. Koltun, P. Krähenbühl, Probabilistic two-stage detection, arXiv preprint arXiv:2103.07461 (2021).

Debojyoti Biswas received the Ph.D. degree in Computer Science from Texas State University, San Marcos, TX, USA. He is currently a postdoctoral researcher at the Larson Transportation Institute at Penn State University. He received his B.Sc. in Information and Communication Engineering from Noakhali Science and Technology University, Bangladesh. He served as a Lecturer in the Department of Computer Science at Leading University, Bangladesh, from 2019 to 2021, before commencing his Ph.D. studies. He is a peer reviewer for several prestigious journals, including IEEE TGRS, IEEE JSTARS, Springer PFGE, Elsevier ESA, and others. His research interests include computer vision, video understanding, intelligent transportation systems, object detection, and domain adaptation.

Andrew Scouten is a graduate student pursuing an M.S. Degree in Computer Science at Texas State University. He began his research in Computer Vision in early 2023 during his undergraduate study at Texas State, where he contributed to the development of the AI systems to detect bacterial adhesion and corrosion in SEM images from NASA's International Space Station. He is currently working as a research assistant on projects funded by the Texas Department of Transportation, focusing on the application of artificial intelligence to pavement distress detection and measurement. Andrew's research interests include the interdisciplinary applications of AI and deep learning, as well as the explanation and improvement of existing AI systems.



Haitao Gong, Ph.D. is a postdoctoral scholar at Texas State University, where he also earned his Ph.D. in Materials Science, Engineering, and Commercialization. He began his academic journey with a bachelor's degree in Transportation from Huazhong University of Science and Technology, followed by a master's degree in Transportation Planning and Management from Tongji University in Shanghai. Before his doctoral studies, he worked as a transportation engineer at China Communications Construction Company, with a focus on transportation management and planning. His current research interests include pavement management, deep learning, and image processing, with a particular emphasis on AI-driven approaches to infrastructure evaluation.

Feng Wang, PE, Ph.D. is a Professor in Transportation Engineering and Program Coordinator of the Civil Engineering Program of the Ingram School of Engineering at Texas State University in San Marcos, Texas. He also serves as Associate Director of the US DOT Tier I University Transportation Center, CREATE. Feng received his Ph.D. in Transportation Engineering from the University of Texas at Austin. His current research activities include the development of AI/ML-based image-processing methods for automated pavement condition evaluation using 2D/3D pavement surface images, image-based analysis of pavement condition data, and decision-support modeling for pavement management systems and pavement warranty programs. He has received research funding from the US DOT, FHWA, NSF, Texas DOT, and Mississippi DOT.

Jelena Tešić, Ph.D. is an Associate Professor at Texas State University. Before joining academia, she was a research scientist at Mayachitra (CA) and the IBM Watson Research Center (NY). She received her Ph.D. (2004) and M.Sc. (1999) in Electrical and Computer Engineering from the University of California, Santa Barbara, CA, USA, and Dipl. Ing. (1998) in Electrical Engineering from the University of Belgrade, Serbia. Her research focuses on advancing AI for scientific applications, including large-scale object localization and identification in overhead imagery. NSF, NIH, NAVAIR, DARPA, ONR, USDA, and DoE funded Dr. Tešić work. Dr. Tešić is in the organizing committee of the IEEE ICMR 2027, and she has served as the Area Chair for the IEEE ICIP, IEEE ICME, ACM IMX, and ACM MM area chair, and as the Guest Editor for IEEE Multimedia Magazine. She has authored more than 80 peer-reviewed scientific papers and holds seven U.S. patents. She is an IEEE and NAI Senior member.